

# $SU(n) / SO(n)$ : Landweber's Manifolds

Bob Stong — notes for the December 7, 2006 topology seminar.

Transcribed from eight-page handwritten notes. [Download the printable transcript \(PDF\)](#). Page divisions follow the scan; formulas are provided as MathML. The original statements, including its proposed formula on the last page, are retained.

## Page 1: The manifolds and their cohomology

Recently, Peter Landweber wrote to ask some questions about the manifolds  $SU(n) / SO(n)$ . These are quite interesting manifolds and I thought I would tell you a bit about them. At the same time, I will include  $U(n) / O(n)$ .

To begin, one has fibrations

$$SU(n) / SO(n) \xrightarrow{i} BSO(n) \xrightarrow{\otimes \mathbb{C}} BSU(n),$$

$$U(n) / O(n) \xrightarrow{i} BO(n) \xrightarrow{\otimes \mathbb{C}} BU(n).$$

This gives  $n$ -plane bundles  $E_n$  over  $U(n) / O(n)$  and  $SU(n) / SO(n)$  with  $E_n \otimes \mathbb{C}$  being a trivial complex bundle. These are the universal (oriented)  $n$ -plane bundles with trivial complexification.

If you consider the space of real  $n$ -dimensional subspaces  $V \subset \mathbb{C}^n$  with the property that  $iV = V^\perp$  is the orthogonal complement of  $V$ , one has what is called the **Lagrangian Grassmannian**:

$$\Lambda_n = U(n) / O(n).$$

The bundle  $E_n$  consists of the pairs: a subspace and a vector in that subspace. The space

$$S\Lambda_n = SU(n) / SO(n)$$

is called the **special Lagrangian Grassmannian**.

Considering mod 2 cohomology for the fibrations, one has

$$H^*(BSO(n)) = \mathbb{Z}_2[w_2, w_3, \dots, w_n] \leftarrow H^*(BSU(n)) = \mathbb{Z}_2[c_2, c_3, \dots, c_n],$$

$$H^*(BO(n)) = \mathbb{Z}_2[w_1, w_2, \dots, w_n] \leftarrow H^*(BU(n)) = \mathbb{Z}_2[c_1, c_2, \dots, c_n].$$

[Transcription note: in the scan, the right-hand space on the second line is labeled  $BO(n)$ ; the fibration immediately above it has  $BU(n)$  in this position.]

One has

$$(\otimes \mathbb{C})^*(c_i) = w_i^2,$$

since

$$c(E \otimes \mathbb{C}) = w(E \otimes \mathbb{C}) = w(E \oplus E) = w(E)^2.$$

It follows that

$$H^*(U(n)/O(n)) = \mathbb{Z}_2[w_1, \dots, w_n] / (w_1^2, \dots, w_n^2),$$

$$H^*(SU(n)/SO(n)) = \mathbb{Z}_2[w_2, \dots, w_n] / (w_2^2, \dots, w_n^2),$$

given as exterior algebras on the Stiefel–Whitney classes of the bundles  $E_n$ .

One has

$$\dim U(n)/O(n) = 1 + 2 + \dots + n = \frac{n(n+1)}{2} = \binom{n+1}{2},$$

$$\dim SU(n)/SO(n) = 2 + \dots + n = \binom{n+1}{2} - 1.$$

## Page 2: The tangent bundle and the five-dimensional example

One also has  $\pi_1(U(n)/O(n)) = \mathbb{Z}$  and  $\pi_1(SU(n)/SO(n)) = 0$ , so  $SU(n)/SO(n)$  is simply connected.

An interesting result is:

**Fact (Eugenio Calabi).**  $SU(3)/SO(3)$  is a simply connected, nonbounding, 5-dimensional manifold.

**Note.** The cobordism group  $\mathfrak{N}_5 = \mathbb{Z}_2$ , so there is a unique cobordism class of nonbounding 5-manifolds. Thom observed that the generator was a manifold described by Wu, which is actually the Dold manifold  $P(1, 2)$ . An oriented manifold has  $w_1 = 0$ , and I think Wu’s manifold was the first example of an oriented manifold having a nonzero odd-dimensional Stiefel–Whitney class.

**Proof.** The mod 2 cohomology is as follows.

### Mod 2 cohomology of $SU(3)/SO(3)$

| Degree $i$ | 0              | 1 | 2              | 3              | 4 | 5              |
|------------|----------------|---|----------------|----------------|---|----------------|
| $H^i$      | $\mathbb{Z}_2$ | 0 | $\mathbb{Z}_2$ | $\mathbb{Z}_2$ | 0 | $\mathbb{Z}_2$ |
| Generator  | 1              | — | $w_2$          | $w_3$          | — | $w_2 w_3$      |

By Wu's formula,  $\text{Sq}^2 w_3 = w_2 w_3$ , so the Wu class  $v_2 = w_2$ . Then  $v = 1 + w_2$  and

$$\tilde{w} = w(SU(3)/SO(3)) = \text{Sq } v = 1 + w_2 + w_3,$$

so  $\tilde{w}_2 \tilde{w}_3 [SU(3)/SO(3)] \neq 0$ . This proves the fact.

**Note.** To avoid confusion, I will let  $\tilde{w}$  denote the Stiefel–Whitney class of the tangent bundle, using  $w = w(E_n)$ .

**Fact.** The complexification of the tangent bundle is trivial for both  $U(n)/O(n)$  and  $SU(n)/SO(n)$ .

**Proof.** The tangent bundle of  $U(n)/O(n)$  is known to be  $S^2(E_n)$ , the second symmetric power of  $E_n$ . Then

$$S^2(E_n) \otimes \mathbb{C} = S^2(E_n \otimes \mathbb{C}) = S^2(\text{trivial complex bundle}) = \text{trivial bundle}.$$

The inclusion  $SU(n)/SO(n) \subset U(n)/O(n)$  is a codimension-one submanifold. Being simply connected, its normal line bundle is trivial. Thus for  $SU(n)/SO(n)$ ,

$$\tau + 1 = S^2(E_n),$$

and the complexification  $\tau \otimes \mathbb{C} + 1_{\mathbb{C}}$  is trivial. Being in the stable range for complex vector bundles,  $\tau \otimes \mathbb{C}$  is also trivial.

## Page 3: Cobordism and symmetric unitary matrices

**Note.** For a manifold  $M^N$  with  $\tau \otimes \mathbb{C}$  trivial, Gromov and Lees have shown that there is a Lagrangian immersion  $\varphi : M^N \rightarrow \mathbb{C}^N$ ; that is, an immersion with

$$i\varphi_*\tau_p M = (\varphi_*\tau_p M)^\perp.$$

This is also totally real:

$$\varphi_*\tau_p M \cap i\varphi_*\tau_p M = \{0\}.$$

Audin has shown that  $U(n)/O(n)$  has a Lagrangian embedding of  $M^N$  in  $\mathbb{C}^N$ . Peter is interested in knowing whether  $SU(n)/SO(n)$  has a totally real embedding.

Once upon a time, Larry Smith and I calculated the cobordism group for manifolds for which the complexification of the stable tangent bundle was trivial. Our result was that the generators of

$$\mathfrak{N}_* = \mathbb{Z}_2[x_i \mid i \neq 2^s - 1]$$

can be chosen so that

$$\Omega_*^{U/O} = \mathbb{Z}_2[x_i \mid i \text{ odd}, i \neq 2^s - 1].$$

Calabi's observation then shows that  $SU(3)/SO(3)$  is the 5-dimensional generator, and it would be interesting to know the cobordism classes of the other manifolds.

**Fact.**

$$SU(n)/SO(n) = \{X \in SU(n) \mid {}^tX = X\},$$

$$U(n)/O(n) = \{X \in U(n) \mid {}^tX = X\}.$$

**Proof.** Consider the function  $U(n) \rightarrow U(n)$  sending  $Y$  to  $Y {}^tY$ , where  ${}^tY$  is the transpose of  $Y$ . This maps onto the symmetric matrices, since

$${}^t(Y {}^tY) = {}^t({}^tY) {}^tY = Y {}^tY,$$

and sends  $U(n)/O(n)$  diffeomorphically to the symmetric matrices. Here  $O(n)$  consists of the matrices with  $Y {}^tY = 1$ . This is proved by Mimura and Sugata.

**Corollary.** For  $n$  even,  $SU(n)/SO(n)$  bounds, and  $U(n)/O(n)$  is always a boundary.

**Proof.** For  $n$  even,  $-1 \in SU(n)$  is a central element, and multiplication by  $-1$  is a free involution on  $\{X \in SU(n) \mid {}^tX = X\}$ . For every  $n$ , a diagonal matrix with diagonal entry  $z \in S^1$  is a central element of  $U(n)$ , and multiplication by these matrices defines a free circle action on  $\{X \in U(n) \mid {}^tX = X\}$ .

**Note.**  $SU(2)/SO(2) = S^3/S^1 = S^2$ , and the involution just described is the generalization of the antipodal involution. The space  $SU(1)/SO(1)$  is a point and is nonbounding;  $SU(1) = SO(1)$  is the unit group.

## Page 4: Exterior powers and Stiefel–Whitney classes

To study the cobordism class of  $SU(n)/SO(n)$  for  $n$  odd, one would like to know the Stiefel–Whitney class

$$\tilde{w} = w(SU(n)/SO(n)) = w(S^2(E_n)).$$

Consider a general  $n$ -plane bundle  $\gamma_n$ . One has

$$\gamma_n \otimes \gamma_n = S^2(\gamma_n) \oplus \Lambda^2(\gamma_n),$$

where  $\Lambda^2$  is the second exterior power, using the decomposition

$$a \otimes b = \frac{1}{2}(a \otimes b + b \otimes a) + \frac{1}{2}(a \otimes b - b \otimes a).$$

Apply the splitting principle to write  $\gamma_n$  as a sum of line bundles  $\ell_1 \oplus \cdots \oplus \ell_n$ , with  $w(\ell_i) = 1 + x_i$ . Then

$$\begin{aligned} w(\gamma_n \otimes \gamma_n) &= \prod_{i,j} (1 + x_i + x_j) \\ &= \prod_i (1 + x_i + x_i) \prod_{i < j} (1 + x_i + x_j) \prod_{i > j} (1 + x_i + x_j) = \left( \prod_{i < j} (1 + x_i + x_j) \right)^2, \end{aligned}$$

and

$$w(\Lambda^2(\gamma_n)) = \prod_{i < j} (1 + x_i + x_j).$$

It follows that

$$w(S^2(\gamma_n)) = w(\Lambda^2(\gamma_n)) = \prod_{i < j} (1 + x_i + x_j).$$

Unfortunately, there is no known formula to describe  $w(\Lambda^2(\gamma_n))$  in terms of  $w(\gamma_n)$ . However, one has:

**Lemma.** For any  $n$ -plane bundle  $\gamma_n$ ,

$$w(\Lambda^s(\gamma_n)) = 1 + \binom{n-1}{s-1} w_1 + \binom{n-2}{s-1} (w_2 + w_3 + \cdots + w_n)$$

modulo decomposables.

**Proof.** Using the splitting principle to write  $w(\gamma_n) = \prod_{i=1}^n (1 + x_i)$ , one has

$$w(\Lambda^s(\gamma_n)) = \prod_{1 \leq i_1 < i_2 < \cdots < i_s \leq n} (1 + x_{i_1} + x_{i_2} + \cdots + x_{i_s}).$$

To establish the formula, it is sufficient to find  $\alpha$  with

$$w_{2^t}(\Lambda^s(\gamma_n)) = \alpha w_{2^t}$$

modulo decomposables. For  $2^t \leq k = 2^t + i < 2^{t+1}$ , working modulo decomposables,

$$w_k(\Lambda^s(\gamma_n)) \equiv \text{Sq}^i w_{2^t}(\Lambda^s(\gamma_n)) \equiv \text{Sq}^i \alpha w_{2^t} \equiv \alpha w_k,$$

since

$$\text{Sq}^i w_j = \binom{j-1}{i} w_{i+j}$$

modulo decomposables, and  $\text{Sq}^i$  takes decomposables to decomposables.

To find  $\tilde{w}_{2^t} = w_{2^t}(\Lambda^s(\gamma_n))$ , consider the bundle

$$\gamma_n = 2^t \ell + (n - 2^t)$$

over  $\mathbb{RP}^\infty$ , where  $\ell$  is the standard line bundle with  $w(\ell) = 1 + x$ . Then

$$w = w(\gamma_n) = (1 + x)^{2^t} = 1 + x^{2^t}.$$

## Page 5: Completion of the lemma and nonbounding manifolds

Thus  $w_{2^t} = x^{2^t}$  and all decomposable classes of degree  $2^t$  are zero. The set of classes  $x_i$  is

$$\{\underbrace{x, \dots, x}_{2^t}, \underbrace{0, \dots, 0}_{n-2^t}\}.$$

Choosing  $s$  of these classes chooses  $p$  of the  $x$ 's and  $s - p$  zeros, for some  $p \leq s$  and  $p \leq 2^t$ . This can be done in  $\binom{2^t}{p} \binom{n-2^t}{s-p}$  ways. Thus

$$\tilde{w} = \prod_p (1 + px)^{\binom{2^t}{p} \binom{n-2^t}{s-p}}.$$

For  $p$  even,  $px = 0$ , so this becomes

$$\tilde{w} = \prod_{p \text{ odd}} (1 + x)^{\binom{2^t}{p} \binom{n-2^t}{s-p}} = (1 + x)^{\sum_{p \text{ odd}} \binom{2^t}{p} \binom{n-2^t}{s-p}},$$

where the product and sum are over odd  $p$  with  $1 \leq p \leq 2^t, s$ .

For  $2^t = 1$ , this is

$$\tilde{w} = (1 + x)^{\binom{n-1}{s-1}} = 1 + \binom{n-1}{s-1} x + \dots,$$

giving the desired result. For  $2^t > 1$  and  $p$  odd,

$$\binom{2^t}{p} = \frac{2^t}{p} \frac{(2^t - 1)!}{(p-1)!(2^t - p)!} = \frac{2^t}{p} \binom{2^t - 1}{p-1}.$$

Hence

$$\begin{aligned} \tilde{w} &= (1 + x)^{2^t \sum_{p \text{ odd}} \frac{1}{p} \binom{2^t - 1}{p-1} \binom{n-2^t}{s-p}} \\ &= (1 + x^{2^t})^{\sum_{p \text{ odd}} \frac{1}{p} \binom{2^t - 1}{p-1} \binom{n-2^t}{s-p}} \\ &= 1 + \left( \sum_{p \text{ odd}} \frac{1}{p} \binom{2^t - 1}{p-1} \binom{n-2^t}{s-p} \right) x^{2^t} + \dots, \end{aligned}$$

where the coefficient of  $x^{2^t}$  is taken modulo 2. Now, modulo 2,

$$\binom{2^t - 2}{p-1} = \binom{2^t - 1}{p-1} \quad \text{for } p \text{ odd},$$

and  $\binom{2^t - 2}{p-1} = 0$  for  $p$  even. The coefficient therefore becomes

$$\sum_p \binom{2^t - 2}{p-1} \binom{n-2^t}{s-p} = \binom{n-2}{s-1},$$

where the sum is over all  $p$ . This gives the desired result.

**Corollary.** For  $n$  odd,  $SU(n)/SO(n)$  is always nonbounding. In particular,

$$\tilde{w}_2 \tilde{w}_3 \cdots \tilde{w}_n [SU(n) / SO(n)] \neq 0.$$

**Proof.** For  $n$  odd and  $s = 2$ ,  $\binom{n-1}{s-1} = 0$  and  $\binom{n-2}{s-1} \neq 0$  modulo 2, so

$$\tilde{w} = 1 + w_2 + \cdots + w_n$$

modulo decomposables. Then

$$\tilde{w}_2 \tilde{w}_3 \cdots \tilde{w}_n = w_2 w_3 \cdots w_n + \text{terms having more than } n - 1 \text{ factors.}$$

Every product of  $n$  factors in  $H^*(SU(n) / SO(n); \mathbb{Z}_2)$  is zero, so

$$\tilde{w}_2 \tilde{w}_3 \cdots \tilde{w}_n = w_2 w_3 \cdots w_n,$$

the nonzero class of top degree.

**Fact.**  $SU(n) / SO(n)$  is indecomposable in  $\mathfrak{N}_*$  only for  $n = 3$ .

## Page 6: Proof of indecomposability only in dimension five

**Proof.** A  $d$ -dimensional manifold  $M^d$  is indecomposable if and only if the characteristic number  $s_d[M^d] \neq 0$ , where, writing  $w(M) = \prod_i (1 + x_i)$ , one has  $s_d = \sum_i x_i^d$ .

Let  $M^d = SU(n) / SO(n)$ , where  $d = \binom{n+1}{2} - 1$ . Since  $s_{2j} = s_j^2$  and squares are zero in  $H^*(SU(n) / SO(n))$ , the manifold is decomposable if  $d$  is even.

For any bundle, the total  $s$ -class  $s = s_1 + s_2 + s_3 + \cdots$  is equal to

$$\frac{w_{\text{odd}}}{w} = \frac{w_1 + w_3 + w_5 + \cdots}{1 + w_1 + w_2 + w_3 + \cdots}.$$

Since  $\tilde{w}^2 = 1$  in  $H^*(SU(n) / SO(n))$ ,

$$s = \tilde{w}_{\text{odd}} \tilde{w} = \tilde{w}_{\text{odd}} \tilde{w}_{\text{even}} + \tilde{w}_{\text{odd}}^2 = \tilde{w}_{\text{odd}} \tilde{w}_{\text{even}},$$

as  $\tilde{w}_{\text{odd}}^2 = 0$ .

One then has, for  $d = 4p + 1$ ,

$$\begin{aligned} s_d &= (\tilde{w}_0 \tilde{w}_{4p+1} + \tilde{w}_1 \tilde{w}_{4p}) + (\tilde{w}_2 \tilde{w}_{4p-1} + \tilde{w}_3 \tilde{w}_{4p-2}) + \cdots \\ &\quad + (\tilde{w}_{2p-2} \tilde{w}_{2p+3} + \tilde{w}_{2p-1} \tilde{w}_{2p+2}) + \tilde{w}_{2p} \tilde{w}_{2p+1}, \end{aligned}$$

and, for  $d = 4p + 3$ ,

$$\begin{aligned} s_d &= (\tilde{w}_0 \tilde{w}_{4p+3} + \tilde{w}_1 \tilde{w}_{4p+2}) + (\tilde{w}_2 \tilde{w}_{4p+1} + \tilde{w}_3 \tilde{w}_{4p}) + \cdots \\ &\quad + (\tilde{w}_{2p} \tilde{w}_{2p+3} + \tilde{w}_{2p+1} \tilde{w}_{2p+2}). \end{aligned}$$

Also,

$$\begin{aligned}
\text{Sq}^1(\tilde{w}_{2i}\tilde{w}_{2j}) &= \text{Sq}^1\tilde{w}_{2i}\tilde{w}_{2j} + \tilde{w}_{2i}\text{Sq}^1\tilde{w}_{2j} \\
&= (\tilde{w}_{2i+1} + \tilde{w}_1\tilde{w}_{2i})\tilde{w}_{2j} + \tilde{w}_{2i}(\tilde{w}_{2j+1} + \tilde{w}_1\tilde{w}_{2j}) \\
&= \tilde{w}_{2i+1}\tilde{w}_{2j} + \tilde{w}_{2i}\tilde{w}_{2j+1}.
\end{aligned}$$

For  $2i + 2j + 1 = d$ , this equals

$$v_1\tilde{w}_{2i}\tilde{w}_{2j} = \tilde{w}_1\tilde{w}_{2i}\tilde{w}_{2j} = 0,$$

since  $\tilde{w}_1 = 0$ . Thus  $s_d[M^d] = 0$  if  $d = 4p + 3$ . If  $d = 4p + 1$ ,

$$s_d[M^d] = \tilde{w}_{2p}\tilde{w}_{2p+1}[M^d] = \tilde{w}_{2p}\text{Sq}^1\tilde{w}_{2p}[M^d].$$

If  $\tilde{w}_{2p}$  is decomposable, write it as a sum of monomials  $\sum z_{i_1} \cdots z_{i_r}$ . Expanding gives

$$\tilde{w}_{2p}\text{Sq}^1\tilde{w}_{2p} = \sum_{I,J} (z_{i_1} \cdots z_{i_r}) \text{Sq}^1(z_{j_1} \cdots z_{j_s}).$$

For distinct monomials, the two cross terms add to

$$\text{Sq}^1(z_{i_1} \cdots z_{i_r} z_{j_1} \cdots z_{j_s}) = v_1(z_{i_1} \cdots z_{i_r} z_{j_1} \cdots z_{j_s}) = 0.$$

Only the diagonal terms remain:

$$\begin{aligned}
&\sum_I (z_{i_1} \cdots z_{i_r}) \text{Sq}^1(z_{i_1} \cdots z_{i_r}) \\
&= \sum_I (z_{i_1} \cdots z_{i_r}) \sum_k z_{i_1} \cdots (\text{Sq}^1 z_{i_k}) \cdots z_{i_r}.
\end{aligned}$$

Every term is zero because it has a factor  $z_i^2$ . Thus  $s_d[M^d] = 0$  if  $\tilde{w}_{2p}$  is decomposable.

For  $M^d = SU(n)/SO(n)$ , the manifold bounds when  $n$  is even. For  $n = 2q + 1$  odd,  $\tilde{w}_{2j}$  is decomposable for  $2j > 2q$ . Thus if  $s_d[M^d] \neq 0$ , then

$$d = 2 + 3 + \cdots + 2q + (2q + 1) \leq 2q + 2q + 1,$$

and  $2q + 1 = 3$ . This proves the fact.

## Page 7: Summary and further formulas

Combining the results:

**Fact.**  $SU(n)/SO(n)$  bounds if  $n$  is even and is nonbounding for  $n$  odd. Also,  $SU(n)/SO(n)$  is indecomposable only for  $n = 3$ .

**Note.** To prove  $s = w_{\text{odd}}/w$ , one has

$$\begin{aligned}
& \frac{w_{\text{odd}}(E + F)}{w(E + F)} \\
= & \frac{w_{\text{odd}}(E)w_{\text{even}}(F) + w_{\text{odd}}(E)w_{\text{odd}}(F) + w_{\text{odd}}(E)w_{\text{odd}}(F) + w_{\text{even}}(E)w_{\text{odd}}(F)}{w(E)w(F)} \\
& = \frac{w_{\text{odd}}(E)w(F)}{w(E)w(F)} + \frac{w_{\text{odd}}(F)w(E)}{w(F)w(E)} \\
& = \frac{w_{\text{odd}}(E)}{w(E)} + \frac{w_{\text{odd}}(F)}{w(F)}.
\end{aligned}$$

If  $L$  is a line bundle with  $w(L) = 1 + x$ , then

$$\frac{w_{\text{odd}}(L)}{w(L)} = \frac{x}{1 + x} = x + x^2 + x^3 + \cdots = s(L).$$

The result follows from the splitting principle.

**Comment.** One has

$$\begin{aligned}
\Lambda^2(\gamma_n + 2) &= \Lambda^2(\gamma_n) \otimes \Lambda^0(2) + \Lambda^1(\gamma_n) \otimes \Lambda^1(2) + \Lambda^0(\gamma_n) \otimes \Lambda^2(2) \\
&= \Lambda^2(\gamma_n) + 2\gamma_n + 1.
\end{aligned}$$

Under the inclusion of  $SU(n)/SO(n)$  in  $SU(n+2)/SO(n+2)$ , the bundle  $E_{n+2}$  restricts to  $E_n$  (with trivial summands understood). Thus  $w(SU(n+2)/SO(n+2))$  restricts to

$$w(SU(n)/SO(n)) w(E_n)^2 = w(SU(n)/SO(n)).$$

There is therefore a universal formula for  $w(SU(n)/SO(n))$ , depending only on  $n$  modulo 2. Using the formula for  $\Lambda^2(\gamma_n + 1)$  relates the even and odd formulas by

$$w(SU(\text{even})/SO(\text{even})) = w(SU(\text{odd})/SO(\text{odd})) w.$$

**Comment.** One also has a formula

$$w(S^s(\gamma_n)) = 1 + \binom{n+s-1}{s-1} w_1 + \binom{n+s}{s-1} (w_2 + \cdots + w_n)$$

modulo decomposables.

One can also ask for similar formulas for complex vector bundles, working over the integers. It appears that the following holds.

## Page 8: The proposed complex-bundle formula

$$c_k(\Lambda^s(\gamma_n^{\mathbb{C}})) = \left\{ \binom{n}{s-1} - \binom{n}{s-2}2^{k-1} + \binom{n}{s-3}3^{k-1} + \cdots + (-1)^{s-2} \binom{n}{1}(s-1)^{k-1} + (-1)^{s-1}s^{k-1} \right\} c_k$$

modulo decomposables.

This formula is correct for  $s = 2$  and  $3$ , and for  $k = 1$  and  $2$  (where the coefficients reduce to  $\binom{n-1}{s-1}$  and  $\binom{n-2}{s-1}$ ). It has the correct mod 2 reduction.

For  $s = 2$ , the formula is

$$\begin{aligned} c(\Lambda^2(\gamma_n^{\mathbb{C}})) &= 1 + (n-1)c_1 + (n-2)c_2 + (n-4)c_3 + \cdots \\ &\quad + (n-2^{k-1})c_k + \cdots + (n-2^{n-1})c_n \end{aligned}$$

modulo decomposables. One also has

$$c(S^2(\gamma_n^{\mathbb{C}})) = 1 + (n+1)c_1 + (n+2)c_2 + (n+4)c_3 + \cdots + (n+2^{n-1})c_n$$

modulo decomposables.